Numerical and simulation techniques in finance

Nicolas Langrené

Beijing Normal-Hong Kong Baptist University

2025 Fall Semester

Teachers

Teacher

Nicolas Langrené (尼古拉·拉萼尼)

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Teaching Assistant

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Lectures + tutorials

Lectures: 3 hours per week

▷ see timetable

Lab tutorials: 1 hour per week

▷ time and venue to be discussed with TA

Office hours

Can be accessed from my timetable at https://staff.uic.edu.cn/nicolaslangrene/en

Dr. Nicolas Langrené, Semester 1 of 2025/2026

	Monday	Tuesday	Wednesday	Thursday	Friday
08:00-08:50					
09:00-09:50					
10:00-10:50	Numerical and Simulation Techniques in Finance T8-301	Office Hour			
11:00-11:50	Numerical and Simulation Techniques in Finance T8-301	Office Hour			
12:00-12:50					
13:00-13:50	Office Hour	Office Hour			
14:00-14:50	Office Hour	Office Hour			Numerical and Simulation Techniques in Finance T8-301
15:00-15:50	Office Hour	Office Hour			Office Hour
16:00-16:50	Introduction to Econometrics T4-603	Office Hour			Introduction to Econometrics T4-501
17:00-17:50					Introduction to Econometrics T4-501

Evaluation

- Homework assignments: **15%**
- Group project: **20%**
- In-class exercises/quizzes: **25%**
- Final examination: **40%**

Introduction

Introduction

The aim is to provide an overview of the basic computational tools that are used in financial engineering. The course combines introductions to pricing financial derivatives by finite difference methods and by Monte Carlo simulation, and a brief introduction to more advanced numerical techniques in finance.

The emphasis is on **practical implementations** and **numerical issues**.

Outline (1/2)

Programming

- Introduction to structured programming
- Examples of programs
- Introduction to Python
- Programming with Python

Numerical computation

- Computer arithmetic
- Measuring errors
- Numerical instability and ill conditioning
- Computational complexity

Outline (2/2)

Finite difference methods

- Numerical differentiation
- Partial differential equations
- Black-Scholes PDE
- Numerical stability of PDE schemes

Monte Carlo methods

- Generating random numbers
- Modeling dependencies
- Monte Carlo option pricing
- Variance reduction

References

[1] M. Gilli, D. Maringer, and E. Schumann. *Numerical methods and optimization in finance*. 2nd. Academic Press, 2019.

[2] Y. Hilpisch. *Python for finance: mastering data-driven finance*. 2nd. O'Reilly Media, Inc., 2018.

[3] R. Korn, E. Korn, and G. Kroisandt. *Monte Carlo methods and models in finance and insurance*. CRC press, 2010.

[4] D. Duffy. *Finite difference methods in financial engineering: a partial differential equation approach*. John Wiley & Sons, 2013.

Python programming language

Installation https://www.anaconda.com/products/distribution
NumPy package: numerical computing for matrices and multi-dimensional arrays https://numpy.org/doc/stable/user/absolute_beginners.html
SciPy package: algorithms for optimization, statistics, interpolation, etc.
https://docs.scipy.org/doc/scipy/tutorial/index.html#user-guide
Pandas package: time series data analysis
https://pandas.pydata.org/docs/getting_started/index.html
Matplotlib package: plotting and visualization
https://matplotlib.org/stable/gallery/index
Seaborn package: statistical data visualization
https://seaborn.pydata.org/examples/index.html

Computational finance applications

- financial option pricing
- option replication and hedging
- numerical approximation of derivatives
- financial scenario simulation
- financial model calibration
- optimal portfolio selection
- backtesting trading strategy
- stress-testing portfolio risk